Question 1 – Car Purchase History Analysis (20%)

Write a module to read the purchase history file (i.e., *car_sales.txt*); then count and display the most frequent **FIVE** car brands that being sold in the log. The results **MUST** contain the "*top-5 frequent Car Brands and their sales count*". You can return the results in any format (e.g., list, dict), not necessarily in dataframe. E.g., (*Car Brand*, *count*)

```
+----------+----------+
| Chevrolet|       177|
|     Dodge|       173|
|      Ford|       167|
|Mitsubishi|       146|
|Volkswagen|       131|
+----------+----------+
```

The purchase history file that we use for this assignment is the car sales file. The file looks something like the following: Ford - - 2015 SUV Auto Black 26

Each part of this file is described below.

- Ford: This is the car brand of one sales record, which means that for this sales record, the car that being sold is a Ford car. (**This is what you need to extract**!)
- - : The "hyphen" in the output indicates that the requested piece of information (user identity) is not available.
- - : The "hyphen" in the output indicates that the requested piece of information (user DOB) is not available.
- 2015 : The time that the car was being sold.
- SUV : This is the car type of the sale record, we have "SUV", "Passenger", "Hardtop" and so on.
- Auto : This means the car is in "Auto mode" or in "Manual mode".
- Black : This is the color of the car.
- 26 : This is the sales price of the car (26 means 26,000$).

In general, your module will contain (but not limited to) the following parts:

1. A component that reads the log file
2. A function that extracts the car brand (e.g., Ford, Chevrolet, …) from the file (**Hints**: the car brands are ***the starting string to the first "-" of each row***. You may use ***slicing*** to extract the substring from text (e.g., link); and use *map()* to apply *slicing* on each row of the text file)
3. A component that counts the car sales amount and sort them descendingly.

Question 2

a) Construct a Student Table as a PySpark DataFrame containing relevant student information as below. (The content is provided in the file student_data.txt, which can be found on Blackboard) (10%)

StudentID	Name	Age	Gender	Grade	Year of Study	Major	Homework	Past GPA
12aB34	Alice	20	F	B+	2	CS	The sky is blue. I enjoy reading books. Cats are curious.	3.5
44rT32	Daniel	20	M	B	2	CS	Cats are curious. The sky is blue. I enjoy reading books.	3.4
05Xx21	Nora	21	F	A	3	CS	Algorithms are efficient. Data structures organize data. Python is popular.	3.6
98Zx56	Brian	21	M	A-	3	Lit	I enjoy reading books. Cats are curious. The sky is blue.	3.7
75qQ12	Faisal	23	M	C+	5	Lit	Shakespeare wrote plays. Poetry uses metaphors. Literature reflects society.	2.9
67Lm89	Carla	22	F	A	4	Bio	Photosynthesis needs sunlight. Leaves are green. Plants grow in soil.	3.9
88Hg45	Grace	21	F	C	3	Bio	Leaves are green. Plants grow in soil. Photosynthesis needs sunlight.	1.9
310o76	Emily	19	F	A-	1	Hist	The past shapes the future. Wars affect nations. People write history.	3.8
22Ws90	Henry	20	M	B-	2	Hist	Trade routes shaped empires. The Renaissance sparked ideas. Monarchs ruled nations.	3.2
59Kk38	Jack	22	M	B	4	EnvSci	Climate change affects weather. Ecosystems are fragile. We must reduce emissions.	3.3

```
# Define schema
schema = StructType([
    …
])

# Define 10 rows of data with mixed homework similarities
data = [
    …
]

# Create DataFrame
…

# Show the DataFrame
…
```

Then, export the DataFrame to a **JSON file**. (**Note:** You will need to also submit the exported JSON file along with your assignment, refer to *Submission Instructions*).

b) Compute Similarity Scores Among Students of the Same Major. (30%)
For students within the **same major**, compare their homework submissions and calculate a **similarity score** based on ROUGE-L metric. Below are the ROUGE package.

```
!pip install rouge_score
from rouge_score import rouge_scorer
metrics = ['rougeL']
scorer = rouge_scorer.RougeScorer(metrics, use_stemmer=True)
```

Steps:

i. For each student A, compare every sentence in A's homework with each sentence in student B's homework (where B has the same major as A) using **ROUGE-L** similarity.

ii. For each student, display the similarity scores with all other students (within the same major), including:

- `sim_names`: Names of students compared.
- `sim_scores`: Corresponding similarity scores.

iii. Construct the result DataFrame with the following columns: *"Name", "Age", "Gender", "Grade", "Year of Study", "Major", "Homework", "Past GPA", "sim_names", "sim_scores"*

iv. Sort the final output by **Major**, then by **Name**.

Name	Age	Gender	Grade	Year of Study	Major	Homework	Past GPA	sim_names	sim_scores
Carla	22	F	A	4	Bio	Photosynthesis needs sunlight. Leaves are green. Plants grow in soil.	3.9	[Grace]	[3.0]
Grace	21	F	C	3	Bio	Leaves are green. Plants grow in soil. Photosynthesis needs sunlight.	1.9	[Carla]	[3.0]
Alice	20	F	B+	2	CS	The sky is blue. I enjoy reading books. Cats are curious.	3.5	[Daniel, Nora]	[3.0, 0.619]
Daniel	20	M	B	2	CS	Cats are curious. The sky is blue. I enjoy reading books.	3.4	[Alice, Nora]	[3.0, 0.619]
Nora	21	F	A	3	CS	Algorithms are efficient. Data structures organize data. Python is popular.	3.6	[Alice, Daniel]	[0.619, 0.619]
Jack	22	M	B	4	EnvSci	Climate change affects weather. Ecosystems are fragile. We must reduce emissions.	3.3	[]	[]
Emily	19	F	A-	1	Hist	The past shapes the future. Wars affect nations. People write history.	3.8	[Henry]	[0.5556]
Henry	20	M	B-	2	Hist	Trade routes shaped empires. The Renaissance sparked ideas. Monarchs ruled nations.	3.2	[Emily]	[0.7778]
Brian	21	M	A-	3	Lit	I enjoy reading books. Cats are curious. The sky is blue.	3.7	[Faisal]	[0.0]
Faisal	23	M	C+	5	Lit	Shakespeare wrote plays. Poetry uses metaphors. Literature reflects society.	2.9	[Brian]	[0.0]

c) Identify Plagiarism Cases: (20%)

Assume that a **similarity score of 3** (i.e., sim_scores = 3) indicates plagiarism between students of the **same major**.

Implement a DataFrame querying system to answer the following questions:

- **"Who plagiarized in Major X?" (x mean a particular Major, e.g., Bio)**
- **"How many students plagiarized in each major?"**
- **Notes:**
  (15%) will be awarded for correctly implementing the query system.
  (5%) will be awarded if the system can handle **different phrasings or patterns** of similar questions
  (e.g., "Who copied in Major X?", "List students who cheated in CS").

Your system should return clear and concise results (in Dataframe) that answer the questions above. E.g.,

```
Question: Who plagiarized in the Major bio?
Answer:
+---------+---------+-----+---------+
|StudentA|StudentB|Major|SimScore|
+---------+---------+-----+---------+
|   Carla|   Grace|  Bio|     3.0|
+---------+---------+-----+---------+

Question: Who plagiarized in the Major EnvSci?
Answer:
+---------+---------+-----+---------+
|StudentA|StudentB|Major|SimScore|
+---------+---------+-----+---------+
+---------+---------+-----+---------+

Question: Who plagiarized in the Major hist?
Answer:
+---------+---------+-----+---------+
|StudentA|StudentB|Major|SimScore|
+---------+---------+-----+---------+
+---------+---------+-----+---------+

Question: How many students plagiarized in each major?
Answer:
+-----+----------------+
|Major|Plagiarism_Count|
+-----+----------------+
|  Bio|               1|
|   CS|               1|
+-----+----------------+
```

d) Determine Copying Relationships Based on GPA (20%)
   - If the GPA difference ≤ 0.3, classify the relationship as "copy each other".
   - If the GPA difference > 0.3, classify as a "copee-copier" relationship:
     - The student with the higher GPA is the copee.
     - The student with the lower GPA is the copier.

StudentA	GPA_A	StudentB	GPA_B	Major	SimScore	Relation
Alice	3.5	Daniel	3.4	CS	3.0	copy each other
Carla	3.9	Grace	1.9	Bio	3.0	copee-copyer

After completing your work, **upload the following files to Blackboard** as instructed by your teacher:

1. **Your Jupyter Notebook file** (containing your full implementation)
2. **The exported Student Table JSON file** (as required in **Question 2a**)

Important Notes (Mark will be deducted without following):

- For this assignment, PySpark should be your primary development tool. While minimal use of standard Python is permissible in exceptional circumstances (e.g., some operators in the function like ROUGE), relying heavily on it will result in point deductions. The vast majority of your code must be written in PySpark.
- The use of OCR (Optical Character Recognition) tools is strictly prohibited for this assignment.
- Plagiarism will be penalized.
- Late submission will be subject to a deduction of **10% per day, not accept any late submission after 3 days**.
- Students who are found to have used GAI-generated work in their coursework/assessments will be committing an act of **academic dishonesty**.
- Students should NOT present GAI-generated work (in part or in whole) as their own, as this is considered a violation of academic integrity.